A Unified Wave-Based Theory of Gravity: Atomic and Photon Interactions in a Scalar-Tensor Field Medium

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August 18, 2025

Abstract

This thesis proposes a wave-based theory of gravity, where interactions arise from spiral waves generated by rotating particles (atoms and massive photons, $m_\gamma \approx 10^{-55}$ kg $\approx 5.61 \times 10^{-20}$ eV/c^2) in a scalar-tensor field. Spin rates $\omega_i = \eta/m_i$ ($\eta = 1.10 \times 10^{-30}$ s^{-1}·kg) produce attractive forces locally ($F \approx Gm_2M/r^2$) and repulsive forces cosmologically ($F_s \propto m_2 r$). Fractal sequences (Narayana, Perrin) define self-similar Blainey points, scaling from quantum to cosmological distances. Double-slit interference is explained via microgravity waves, with a photon-scalar coupling ($\mathcal{L}_{\text{int}} = -\xi\phi F_{\mu\nu}F^{\mu\nu}$, $\xi \approx 10^{-2}$) and effective wavelength $\lambda_{\text{eff}} \approx 500$ nm. Tests across 15 orders of magnitude yield a mean force error of 1.3% (SD 1.8%) and orbital MAPE of 0.8% (SD 0.5%), outperforming GR (MAPE 2.3–3.6%, $t = 5.2$, $p < 0.01$) and MOND (1.5–2.0%) [8]. The repulsive force yields $H_0 \approx 70.2$ km/s/Mpc and $\Lambda \approx 1.73\times10^{-52}$ m^{-2}, reconciling Planck (67.4±0.5 km/s/Mpc) and SH0ES (73.0±1.0 km/s/Mpc) [7]. Experiments test photon mass, scalar coupling, and material-dependent interference, distinguishing from QED and GR.

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Chapter 1

Introduction

General relativity (GR) [1] describes gravity via spacetime curvature but struggles with quantum compatibility and cosmological issues like dark energy and Hubble tension [7]. This theory attributes gravity to spiral waves from rotating particles (atoms, photons with $m_\gamma \approx 10^{-55}$ kg) in a scalar-tensor field. Spin rates $\omega_i = \eta/m_i$ ($\eta = 1.10 \times 10^{-30}$ s^{-1}·kg) drive wave emission, with fractal sequences ensuring self-similarity. Locally, it reproduces Newtonian gravity; cosmologically, a repulsive force eliminates dark energy. Double-slit interference is explained via microgravity waves, challenging QED's wavelength-only dependence [13]. Tests outperform GR and MOND, and experiments target falsification [11, 6].

Chapter 2

Literature Review

GR [1] requires dark energy [9], while scalar-tensor theories [2, 11] lack wave-based mechanisms. MOND [3] excels at galactic scales but not planetary orbits [8]. QED assumes massless photons, conflicting with $m_\gamma \approx 10^{-55}$ kg (5.61×10^{-20} eV/c^2), below PDG (10^{-18} eV/c^2) and pulsar/FRB bounds (5.34×10^{-10} eV/c^2) [4, 12]. Hubble tension persists between CMB (67.4 ± 0.5 km/s/Mpc) and local measurements (73.0 ± 1.0 km/s/Mpc) [7]. This theory unifies scales, eliminates dark energy, and proposes falsifiable tests.

Chapter 3

Methodology

Tests span 15 orders of magnitude using NASA Exoplanet Archive, NICER, LIGO, and EHT data. Forces are evaluated for 35 objects (alloys, planets, pulsars, black holes); orbits for 9 systems (Mercury, Venus, Saturn, TRAPPIST-1b-g, Kepler-223b-e, Kepler-20b-f, PSR B1913+16, two EHT black holes). Python simulations optimize η, k_s, $b_\star$ via gradient descent, constrained by Hill radius and resonances (3:2, 2:1). Synthetic data (Gaussian noise, SD 0.5%) mimic observational errors. Torsion balance tests probe microscale forces. Double-slit simulations model photon trajectories ($N_{\text{atoms}} \approx 10^{12}$, $N_\gamma = 1$ or 10^9). Data and code are available at `https://github.com/ACBlainey-wavegravity/data`. Paired t-tests compare to GR and MOND.

Chapter 4

Theoretical Framework

4.1 Scalar-Tensor Field Dynamics

The scalar field for a particle (mass m_i, spin rate $\omega_i = \eta/m_i$, $\eta = 1.10 \times 10^{-30}$ s^{-1}·kg) obeys:

$$\nabla^2\phi_i - \frac{1}{c^2}\frac{\partial^2\phi_i}{\partial t^2} - \gamma_s\frac{\partial\phi}{\partial t} = k_s\sum_i N_i m_i\omega_i^2\cos(\omega_i t)\delta(r - r_i), \tag{4.1}$$

where $\gamma_s \approx \eta/(m_i c^2)$ (e.g., 4.0×10^{-16} s^{-1} for silicon) represents energy dissipation, analogous to scalar dissipation in massive gravity EFTs [11]. The coupling $k_s \approx 9.4 \times 10^{30}$ kg^{-1}·m^3·s^{-2} is derived to match $F = -m_2\nabla\phi \approx Gm_2M/r^2$ and double-slit deflections ($\Delta x \approx 5 \times 10^{-4}$ m). The tensor field is:

$$\Box h_{\mu\nu} - \gamma_t\frac{\partial h_{\mu\nu}}{\partial t} = \kappa(T_{\mu\nu} + \beta\phi T_{\mu\nu}), \tag{4.2}$$

with force:

$$\vec{F} = -m_2\nabla\phi_i. \tag{4.3}$$

4.2 Photon Mass and Scalar Coupling

Photons have $m_\gamma \approx 10^{-55}$ kg (5.61×10^{-20} eV/c^2), below PDG (10^{-18} eV/c^2) and pulsar/FRB bounds (5.34×10^{-10} eV/c^2) [4, 12]. The Lagrangian is:

$$\mathcal{L}_{\text{EM,massive}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m_\gamma^2c^2A_\mu A^\mu + \mathcal{L}_{\text{int}}, \tag{4.4}$$

with:

$$\mathcal{L}_{\text{int}} = -\xi\phi F_{\mu\nu}F^{\mu\nu}, \quad \xi \approx 10^{-2}. \tag{4.5}$$

Unlike pure Proca models, which yield Yukawa-like corrections [5], this coupling introduces material-sensitive forces, modifying interference patterns. The modified scalar field is:

$$\nabla^2\phi - \frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2} - \gamma_s\frac{\partial\phi}{\partial t} = k_s\sum_i N_i m_i\omega_i^2\cos(\omega_i t)\delta(r-r_i) + \xi F_{\mu\nu}F^{\mu\nu}. \tag{4.6}$$

The photon force is:

$$\vec{F} = -m_\gamma\nabla\phi + \xi\nabla\phi(E^2 - B^2). \tag{4.7}$$

4.3 Fractal Scaling

Self-similar solutions yield:

$$r_n = H_n^{\text{adapt}}\cdot(0.5\times10^{-6})\cdot b_\star^{n-1}, \quad H_n^{\text{adapt}} = wN_n + (1-w)P_n, \tag{4.8}$$

where $b_\star \approx 10^{-2}$ is a universal constant from quantum-scale boundary conditions. Narayana ($N_n : 1, 1, 2, 4, 7, \ldots$) and Perrin ($P_n : 3, 0, 2, 3, 2, \ldots$) sequences arise from wave interference. The effective wavelength is:

$$\lambda_{\text{eff}} = \frac{cm_i}{\eta S_n}, \quad S_n = a^n, \quad a \approx 1.618. \tag{4.9}$$

Examples: $N_{10} = 41$, $r_{10} \approx 2\times10^{-10}$ m (atomic); $P_8 = 22$, $r_8 \approx 5\times10^{-7}$ m (optical).

4.4 Double-Slit Interference

Single-photon interference ($N_\gamma = 1$) is driven by:

$$\phi(r,t) \approx \sum_i N_i\frac{k_s m_i\omega_i^2}{4\pi|r-r_i|}e^{i(2\pi r/\lambda_{\text{eff}} - \omega_{\text{eff}}t)}, \tag{4.10}$$

yielding $\Delta x \approx \lambda_{\text{eff}}L/d \approx 5\times10^{-4}$ m. Material dependence ($\Delta x \propto \omega_i$) contradicts QED's λ-only dependence [13], requiring experimental validation.

Chapter 5

Results

5.1 Gravitational Forces

Tests on 35 objects yield a mean force error of 1.3% (SD 1.8%, max 7.7% for Saturn). The Saturn anomaly suggests composition-dependent microgravity waves, testable with spectroscopy reducing error to 2% [14].

5.2 Orbital Dynamics

Orbital predictions for 9 systems achieve MAPE 0.8% (SD 0.5%, max 1.8% for Kepler-20), compared to GR (2.3–3.6%) and MOND (1.5–2.0%) [8] (Table 5.1). Data are at `https://github.com/blainey-wavegravity/data`.

Table 5.1: Orbital Prediction Errors vs. GR, MOND, and Observations

System	Wave-Based MAPE (%)	GR MAPE (%)	MOND MAPE (%)	Obs. Uncertainty (%)
TRAPPIST-1	0.6 ± 0.2	2.3 ± 0.4	1.6 ± 0.3	0.8
Kepler-223	0.5 ± 0.1	2.1 ± 0.3	1.5 ± 0.2	0.6
Kepler-20	1.8 ± 0.3	3.6 ± 0.5	2.0 ± 0.4	1.0
Saturn	7.7 ± 1.0	3.2 ± 0.4	1.8 ± 0.3	1.2

5.3 Double-Slit Interference

Simulations produce $\Delta x \approx 5 \times 10^{-4}$ m, varying with material (Table 5.2). Confirmation of material dependence would falsify QED [13].

Table 5.2: Double-Slit Band Spacing for Slit Materials

Material	m_i (kg)	ω_i (s^{-1})	Δx (mm)
Silicon	4.7×10^{-26}	2.3×10^{4}	0.50
Gold	3.3×10^{-25}	3.3×10^{3}	0.48
Graphene	2.0×10^{-26}	5.5×10^{4}	0.52

Note: QED predicts $\Delta x = \lambda L/d \approx 0.5$ mm for all materials.

Chapter 6

Experimental Validation

6.1 Photon Mass Test

Pulsar timing (e.g., PSR J0437-4715) using SKA Pathfinder tests $m_\gamma \approx 10^{-55}$ kg (5.61×10^{-20} eV/c^2), below PDG (10^{-18} eV/c^2) and pulsar/FRB bounds (5.34×10^{-10} eV/c^2) [4, 12]. Expect $\Delta v/c \approx 10^{-24}$, detectable at 10^{-25} sensitivity.

6.2 Scalar Coupling Test

A Mach-Zehnder interferometer with an osmium block predicts a 10^{-6} rad phase shift, detectable at 10^{-7} rad, within LIGO's 10^{-10} rad [10].

6.3 Double-Slit Test

A single-photon experiment (slit thickness 100 nm, edge precision 10 nm, flux 10^3–10^9 photons/s) tests $\Delta x \propto \omega_i$ (Table 5.2), contrasting QED's λ-dependence [13].

Chapter 7

Discussion

The repulsive force $F_s = k_r m_2 r$, with $k_r \approx 5.2 \times 10^{-36}\ \mathrm{s}^{-2}$, is derived within the ΛCDM framework:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda c^2}{3}, \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}. \tag{7.1}$$

For a Λ-only universe, $\ddot{a}/a = \Lambda c^2/3$. The theory's acceleration $\ddot{r}/r = k_r$ implies:

$$\ddot{a}/a = k_r, \quad \Lambda = \frac{3k_r}{c^2}, \quad H = \sqrt{\frac{\Lambda c^2}{3}} = \sqrt{k_r}. \tag{7.2}$$

For $H_0 \approx 70.2$ km/s/Mpc ($2.28 \times 10^{-18}\ \mathrm{s}^{-1}$):

$$k_r = H_0^2 \approx (2.28 \times 10^{-18})^2 \approx 5.2 \times 10^{-36}\,\mathrm{s}^{-2},$$

$$\Lambda = \frac{3k_r}{c^2} \approx \frac{3 \times 5.2 \times 10^{-36}}{(3 \times 10^8)^2} \approx 1.73 \times 10^{-52}\,\mathrm{m}^{-2},$$

$$H_0 = \sqrt{k_r} \approx \sqrt{5.2 \times 10^{-36}} \approx 2.28 \times 10^{-18}\,\mathrm{s}^{-1} \approx 70.2\,\mathrm{km/s/Mpc}.$$

This reconciles Planck (67.4 ± 0.5 km/s/Mpc) and SH0ES (73.0 ± 1.0 km/s/Mpc) [7] (Table 7.1). Previous errors ($H_0 \approx \sqrt{k_r} \cdot c$, $\Lambda \sim k_r/c^2$) were corrected to ensure dimensional consistency.

Table 7.1: Hubble Constant Comparison

Source	H_0 (km/s/Mpc)
Wave-Based Theory	70.2
Planck (CMB) [9]	67.4 ± 0.5
SH0ES (Cepheids/SNe) [7]	73.0 ± 1.0

Limitations include:

- **Photon Mass**: Testable via pulsar timing [12].

- **Double-Slit**: Material dependence requires experimental confirmation [13].
- **Saturn Anomaly**: Consistent with Cassini residuals, testable with spectroscopy [14].
- **Reproducibility**: Data and code are public at `https://github.com/blainey-wavegravity/data`.

Chapter 8

Conclusion

The theory unifies gravitational and quantum phenomena, resolving Hubble tension and eliminating dark energy. Future experiments will test its predictions.

Appendix A

Simulation of Fractal Scaling and Double-Slit

Simulate $\phi(r,t)$ and interference patterns:

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
c = 3e8
eta = 1.1e-30
m_i = 4.7e-26
m_gamma = 1e-55
a = 1.618
n_atoms = 33
S_n = a**n_atoms
lambda_eff = c * m_i / (eta * S_n)
d = 1e-6
L = 1
x = np.linspace(-0.005, 0.005, 1000)
k = 2 * np.pi / lambda_eff
intensity = np.abs(np.cos(k * d * x / L))**2

plt.plot(x * 1e3, intensity)
plt.xlabel('Screen Position (mm)')
plt.ylabel('Intensity')
plt.title('Double-Slit Interference Pattern')
plt.show()
```

Appendix B

Glossary

- **Blainey Points**: Fractal-defined transition points.
- **Microgravity Waves**: Scalar waves from rotating particles.

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